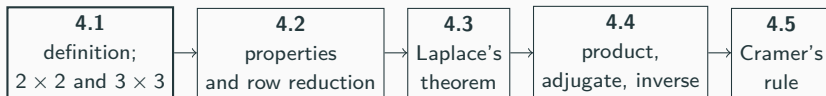


Linear Algebra

4 Determinants

José Firmino Aguilar Madeira



Central idea: each square matrix has an associated number, the determinant, which tells us at once whether the matrix is invertible:

$$A \text{ invertible} \iff \text{car } A = n \iff \det A \neq 0.$$

Exercises: Sheet 4, Ex. 66–82 (original numbering).

4.1 Definition

Determinants of order 2 and 3

Order 2: $\det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = a_{11}a_{22} - a_{12}a_{21},$ for example $\begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix} = 12 - 2 = 10.$

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Order 3 (Sarrus' rule):

$$\det A = (a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}) - (a_{13}a_{22}a_{31} + a_{12}a_{21}a_{33} + a_{11}a_{23}a_{32}).$$

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Example (this course's running matrix):

$$\det C = \begin{vmatrix} 2 & 0 & 0 \\ -6 & 2 & -3 \\ -6 & 0 & -1 \end{vmatrix} = 2 \cdot 2 \cdot (-1) + 0 + 0 - (0 + 0 + 0) = -4.$$

Notation: $\det A$ or $|A|$. Sarrus' rule works *only* for order 3.

General definition

A **term** of A (order n) is a product $a_{1\beta_1} a_{2\beta_2} \cdots a_{n\beta_n}$ with *one* entry from each row and *one* from each column: $(\beta_1, \dots, \beta_n)$ is a permutation of $(1, \dots, n)$.

Definition

The term is *even* or *odd* according as the permutation $(\beta_1, \dots, \beta_n)$ has an even or odd number of **inversions** (pairs out of order). Then

$$\det A = \sum (\text{even terms}) - \sum (\text{odd terms}).$$

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- There are $n!$ terms: 2 for $n = 2$, 6 for $n = 3$ (Sarrus), 24 for $n = 4$, 720 for $n = 6$.
- Swapping two entries of a permutation changes its parity; half of the permutations are even.

For $n \geq 4$, the definition is not a method of computation: we use the properties (4.2) and Laplace (4.3).

- **Ex. 66 a)–e)** determinants of order 2 and 3. In c) and d), first look for a reason why the result is 0.

4.2 Properties and row reduction

Properties

Let A be square of order n . ('line' = row or column)

- I $\det A^T = \det A$: everything said about rows holds for columns.
- II A zero line $\Rightarrow \det A = 0$.
- III Swapping two parallel lines **changes the sign**.
- IV Two equal or proportional parallel lines $\Rightarrow \det A = 0$.
- V Multiplying a line by λ multiplies the determinant by λ . Hence $\det(\lambda A) = \lambda^n \det A$.
- VI Adding to a line a multiple of another parallel line **does not change** the determinant.
- VII If a line is a sum of two, the determinant is the sum of the two corresponding determinants.
- VIII If A is triangular, $\det A = a_{11}a_{22} \cdots a_{nn}$.

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Theorem

A is invertible $\iff \text{car } A = n \iff \det A \neq 0$.

Row-reduction method

Bring the matrix to echelon form with the operations of III, V and VI, recording the effect of each one, and use VIII.

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 11 \end{vmatrix} \xrightarrow{L_2 - 2L_1, L_3 - 3L_1} \begin{vmatrix} 1 & 2 & 3 \\ 0 & -1 & -1 \\ 0 & -1 & 2 \end{vmatrix} \xrightarrow{L_3 - L_2} \begin{vmatrix} 1 & 2 & 3 \\ 0 & -1 & -1 \\ 0 & 0 & 3 \end{vmatrix} = 1 \cdot (-1) \cdot 3 = -3.$$

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What each operation does to the determinant

add a multiple of another line	swap two lines	multiply a line by λ
nothing	$\times(-1)$	$\times\lambda$

It is the same echelon form as in Ch. 3 (rank); now the product of the pivots matters too.

- **Ex. 66 f), g), h)** use the properties before doing any arithmetic.
- **Ex. 67, 72** values of x for which the matrix is singular / non-singular ($\det = 0$ / $\det \neq 0$).
- **Ex. 68–69** compute or compare determinants using *only* the properties.

4.3 Laplace's theorem

Definitions

Let A be of order n .

- M_{ij} : determinant of the submatrix obtained by deleting row i and column j (**minor** of a_{ij}).
- $A_{ij} = (-1)^{i+j} M_{ij}$: **cofactor** of a_{ij} .

The signs $(-1)^{i+j}$ form a chessboard:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & & & \ddots \end{bmatrix}$$

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Laplace's theorem

For any row i : $\det A = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in}$ (and likewise for any column).

Laplace in practice

Choose the line with the most zeros. Along the 1st row of C :

$$\det C = 2 \cdot (+1) \begin{vmatrix} 2 & -3 \\ 0 & -1 \end{vmatrix} + 0 + 0 = 2 \cdot (-2) = -4.$$

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Shortcut method (order ≥ 4): with type VI operations, create zeros in a line until only one non-zero entry is left; expand by Laplace; repeat down to order 3.

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Shortcut method (order ≥ 4): with type VI operations, create zeros in a line until only one non-zero entry is left; expand by Laplace; repeat down to order 3.

Generalised Laplace (extension; in the exercises expanding along one line is enough): expand along m parallel lines, with minors of order m . Along rows 1 and 2 (only columns 2 and 4 give a non-zero minor):

$$\begin{vmatrix} 0 & 2 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 2 & 3 & 4 & 5 \\ 6 & 7 & 1 & 9 \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} \cdot (-1)^{1+2+2+4} \begin{vmatrix} 2 & 4 \\ 6 & 1 \end{vmatrix} = 1 \cdot (-1) \cdot (-22) = 22.$$

- **Ex. 66 f), i), j)** order 4 and 6: row reduction + Laplace.
- **Ex. 71** three determinants of order 4. In a), the 3rd row already has three zeros.

4.4 Product, adjugate and inverse

Theorem

If A and B are square of the same order n : $\det(AB) = \det A \det B$.

Consequences:

$$\det(AB) = \det(BA), \quad \det(A^k) = (\det A)^k, \quad \det(A^{-1}) = \frac{1}{\det A}, \quad \det(\lambda A) = \lambda^n \det A.$$

Determinant of a product

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Caution

$\det(A + B) \neq \det A + \det B$ in general. For example, $\det(I_2 + I_2) = 4 \neq 1 + 1$.

Adjugate matrix and inverse

Definition and theorem

The **adjugate** of A is the transpose of the matrix of cofactors: $\text{adj } A = [A_{ij}]^T$. Then

$$A \text{ adj } A = \text{adj } A A = (\det A) I, \quad \text{and, if } \det A \neq 0 : \quad A^{-1} = \frac{1}{\det A} \text{ adj } A.$$

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Example. For C : $\text{adj } C = \begin{bmatrix} -2 & 0 & 0 \\ 12 & -2 & 6 \\ 12 & 0 & 4 \end{bmatrix}$, $\det C = -4$:

$$C^{-1} = -\frac{1}{4} \text{adj } C = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ -3 & \frac{1}{2} & -\frac{3}{2} \\ -3 & 0 & -1 \end{bmatrix} \quad (\text{the same as by Gauss-Jordan, Ch. 3}).$$

Order 2: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$

- **Ex. 70** determinants of expressions in A , knowing only $\det A$.
- **Ex. 73** inverses via the adjugate.
- **Ex. 74–76** properties of the adjugate. Use $A \operatorname{adj} A = (\det A)I$ and take determinants.

4.5 Cramer's rule

Theorem

If A is square of order n and $\det A \neq 0$, the system $A\mathbf{x} = \mathbf{b}$ is consistent with a unique solution, and

$$x_j = \frac{\Delta_j}{\det A}, \quad \Delta_j = \det(A \text{ with column } j \text{ replaced by } \mathbf{b}).$$

It follows from $\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{\det A} \operatorname{adj} A \mathbf{b}$.

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Example. $C\mathbf{x} = (2, -11, -9)$:

$$\Delta_1 = \begin{vmatrix} 2 & 0 & 0 \\ -11 & 2 & -3 \\ -9 & 0 & -1 \end{vmatrix} = -4, \quad \Delta_2 = -8, \quad \Delta_3 = -12 \Rightarrow \mathbf{x} = \left(\frac{-4}{-4}, \frac{-8}{-4}, \frac{-12}{-4} \right) = (1, 2, 3).$$

Good for one unknown (Ex. 79, 82) or parameters. For big systems, Gaussian elimination is far faster.

- **Ex. 77–78** show that the system is a Cramer system ($\det \neq 0$) and solve it.
- **Ex. 79, 82** only one unknown: a single Δ_j is enough.
- **Ex. 80** order 4.
- **Ex. 81** values of the parameter for which there is a unique solution.

LINEAR ALGEBRA IN PRACTICE: area and volume

The parallelogram spanned by the columns $(3, 1)$ and $(1, 2)$ has area

$$\left| \det \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \right| = |6 - 1| = 5.$$

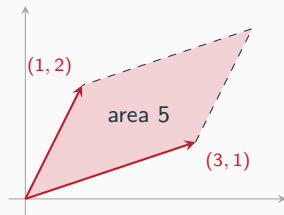
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- In general, $|\det A|$ is the factor by which A multiplies areas (in $\mathbb{R}^2$) or volumes (in $\mathbb{R}^3$).
- $\det A = 0$: the matrix 'flattens' space — there is no volume, and it is not invertible.
- $\det R_\theta = 1$: rotations do not change areas.
- $\det C = -4$: C multiplies volumes by 4; the negative sign indicates that it reverses orientation.

This is why the determinant appears in integral calculus (change of variables).



LINEAR ALGEBRA ON THE COMPUTER: how a determinant is computed

For illustration only — not examinable. Python (NumPy); in MATLAB/Octave: `det(C)`.

```
import numpy as np
C = np.array([[ 2, 0,  0],
              [-6, 2, -3],
              [-6, 0, -1]])
np.linalg.det(C)
# -4.0000000000000001
```

The computer **does not use the definition**: it row-reduces the matrix to triangular form and multiplies the pivots, with the sign of the swaps (4.2).

Why? For $n = 20$, the definition has $20! \approx 2.4 \times 10^{18}$ terms: at 10^9 terms per second, that would take about **77 years**. Row reduction takes about $n^3/3 \approx 2700$ operations.

Caution: because of rounding errors (here, 10^{-15}), we do not test `det(A) == 0`. `matrix_rank` uses a tolerance to estimate the rank numerically.

Quick reference — Chapter 4

Concept	How to get it / what it says
Order 2	$ad - bc$
Order 3	Sarrus' rule (for order 3 only)
Triangular	product of the main diagonal
Operations	add a multiple: nothing; swap: $\times(-1)$; line $\times \lambda$: $\times \lambda$
Zero	zero line, equal or proportional lines $\Rightarrow \det = 0$
Invertibility	A invertible $\iff \text{car } A = n \iff \det A \neq 0$
Laplace	$\det A = \sum_j a_{ij} A_{ij}$, $A_{ij} = (-1)^{i+j} M_{ij}$; choose the line with the most zeros
Product	$\det(AB) = \det A \det B$, $\det A^{-1} = 1/\det A$
Transp., λA	$\det A^T = \det A$, $\det(\lambda A) = \lambda^n \det A$
Adjugate	$\text{adj } A = [A_{ij}]^T$; $A \text{ adj } A = (\det A)I$; $A^{-1} = \text{adj } A / \det A$
Cramer	$x_j = \Delta_j / \det A$ (if $\det A \neq 0$)
Geometry	$ \det A = \text{area / volume factor}$

KEY TAKEAWAYS — Chapter 4

1. The determinant is a number that decides whether a square matrix is invertible: it is non-zero exactly when the matrix is invertible.
2. In practice, to compute determinants of high order, the general definition is not used directly: we row-reduce to a triangular form or expand by Laplace.
3. Adding to a line a multiple of another does not change the determinant; swapping lines changes its sign.
4. The determinant of a product is the product of the determinants, but that of a sum is not the sum.
5. Geometrically, the absolute value of the determinant measures how much the matrix enlarges or shrinks areas and volumes.

Next (Ch. 5): inner product and geometry — lengths, angles, lines and planes.

References (full list: course bibliography)

- Santana, A. P., Queiró, J. F., *Introdução à Álgebra Linear*, Gradiva (in Portuguese).
- Lipschutz, S., Lipson, M., *Schaum's Outline of Linear Algebra*, McGraw-Hill — many solved exercises.
- Strang, G., *Introduction to Linear Algebra*, Wellesley-Cambridge Press.

Resources — Sheet 4 (exercises 66–82) and its solutions.